MINING DEALER BEST PRACTICE

Lug Nut Disassembly and Assembly Support Apparatus

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

February 2022
4.03-220202-1

1.0 Introduction

It's difficult for a single person to perform the removal and tightening process of lug nuts on tires, and there is potential for a safety incident to occur because the tools used could become dislocated and cause injury or damage. Therefore, the current process requires two people - one person to support and the other person to do the removal/tightening process. However, by using this lug nut support apparatus, the tire lug nuts of machines can be removed and tightened in a safer manner with a single person.

2.0 Best Practice Description

The lug nuts on the machine's tires are removed and tightened with a bolt gun. During the removal or tightening of the lug nuts, an extension intermediate arm and counter support are required as they are on the inside of the rim. This process requires two people working together. During removal, some bolts may be jammed and a socket wrench is used with the intermediate arm by applying force. During tightening, the lug nuts must be at torque when fitted. Bolts that are not tightened properly can loosen and even fall. In any of these situations, during removal or tightening, the arm or wrench can dislocate, causing a dangerous situation. Different situations just described can be seen in the pictures below.

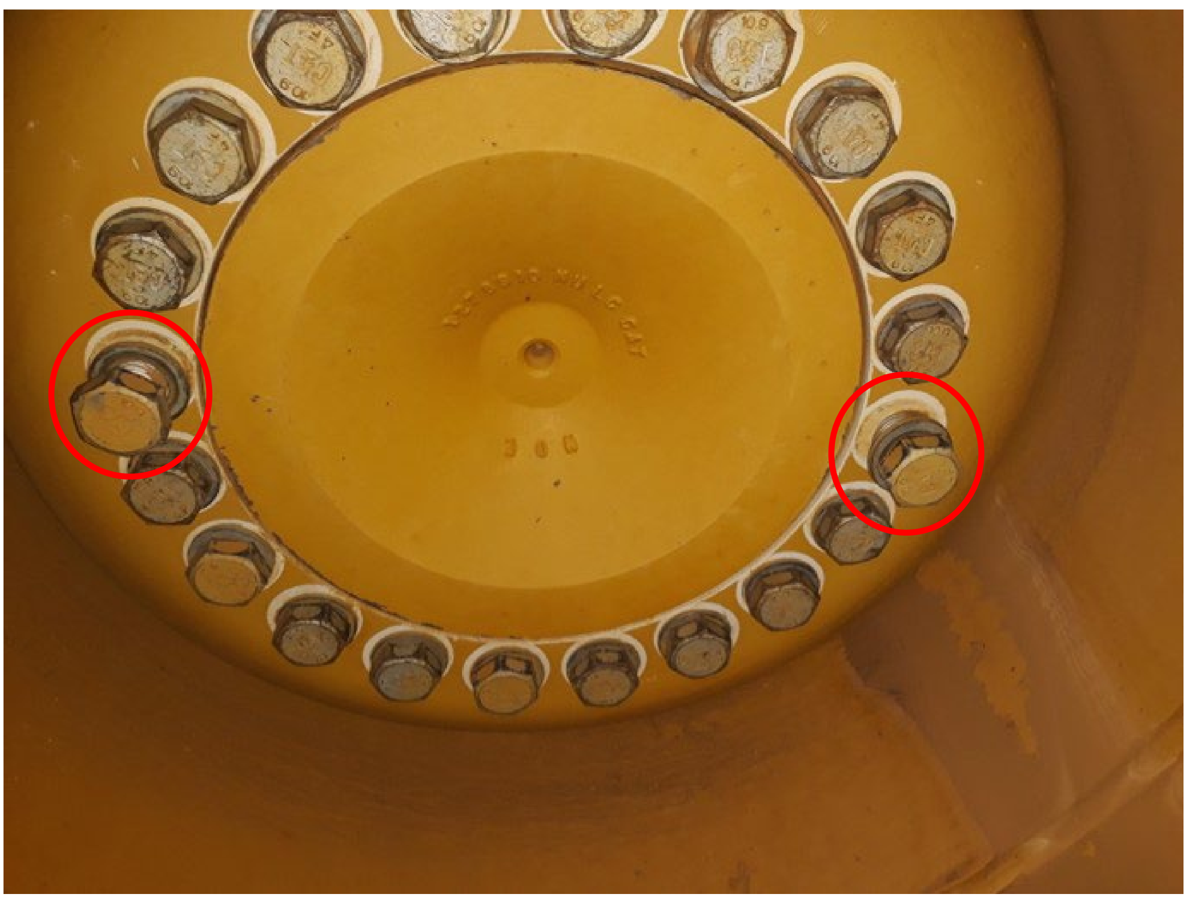

Figure 1: Bolts are loose

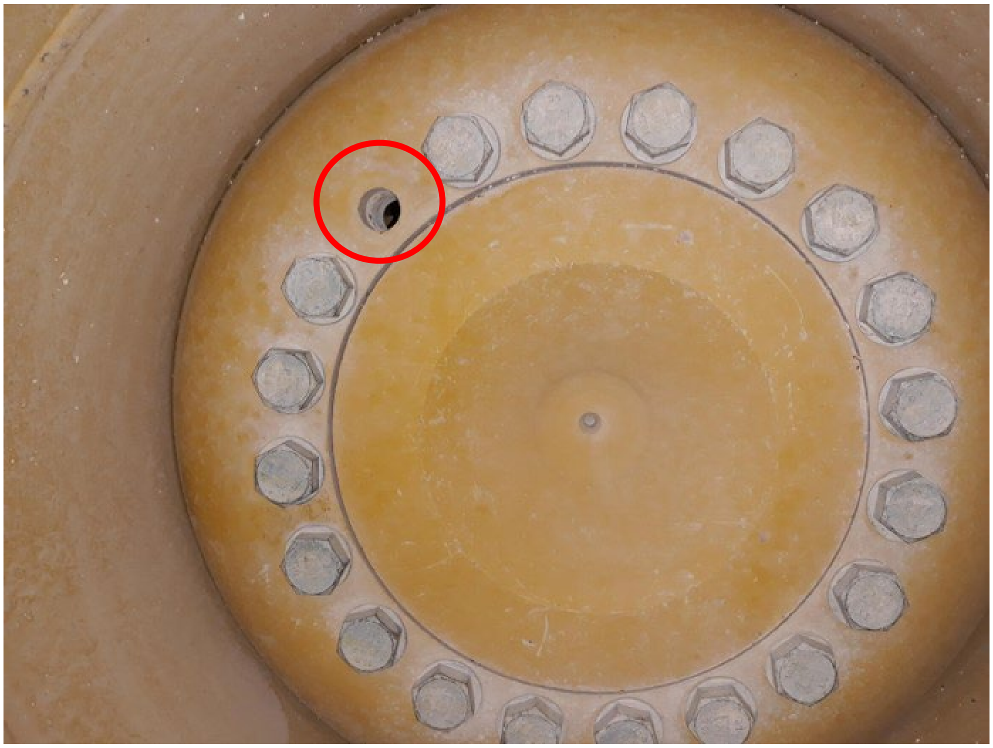

Figure 2: A bolt has fallen

When the apparatus is not used, removal and assembly processes can be seen in the pictures below.

Figure 3: Two-person disassembly of bolts

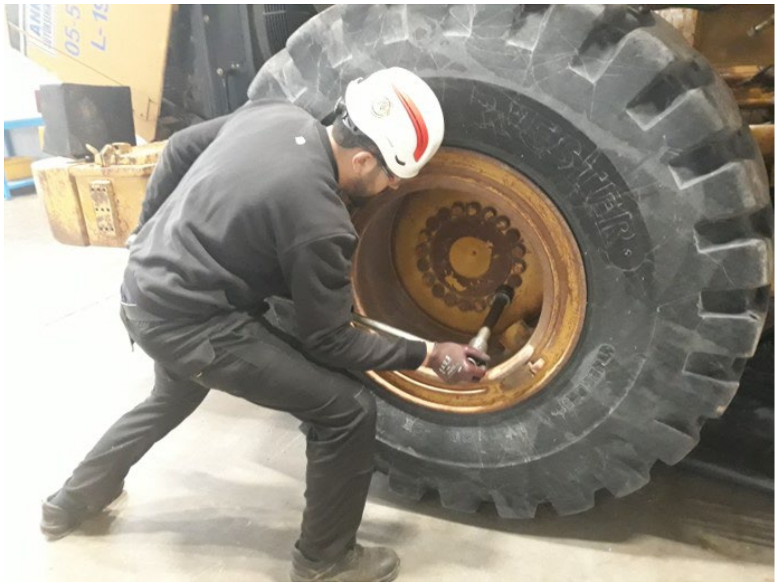

Figure 4: A person slogging on dissembling bolts

3.0 Implementation Steps

1. Tilt the apparatus on its wheels and bring it next to the machine by pushing or pulling it.
2. Adjust the height of the apparatus according to the lug nut height range.
3. Place the long intermediate link in a suitable support place from the various ones attached to the apparatus.
4. Remove or tighten the lug nuts.

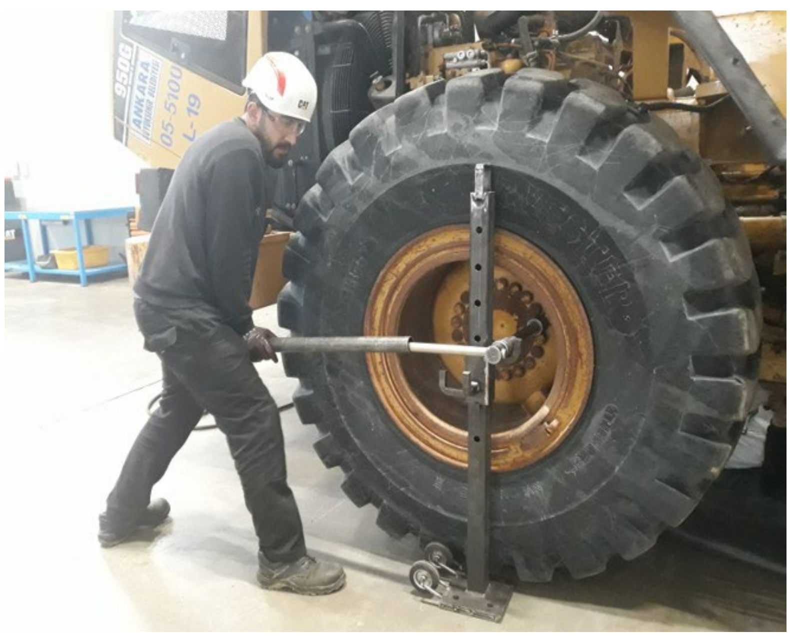

Figure 5: Disassembly of the lug nuts with the new apparatus

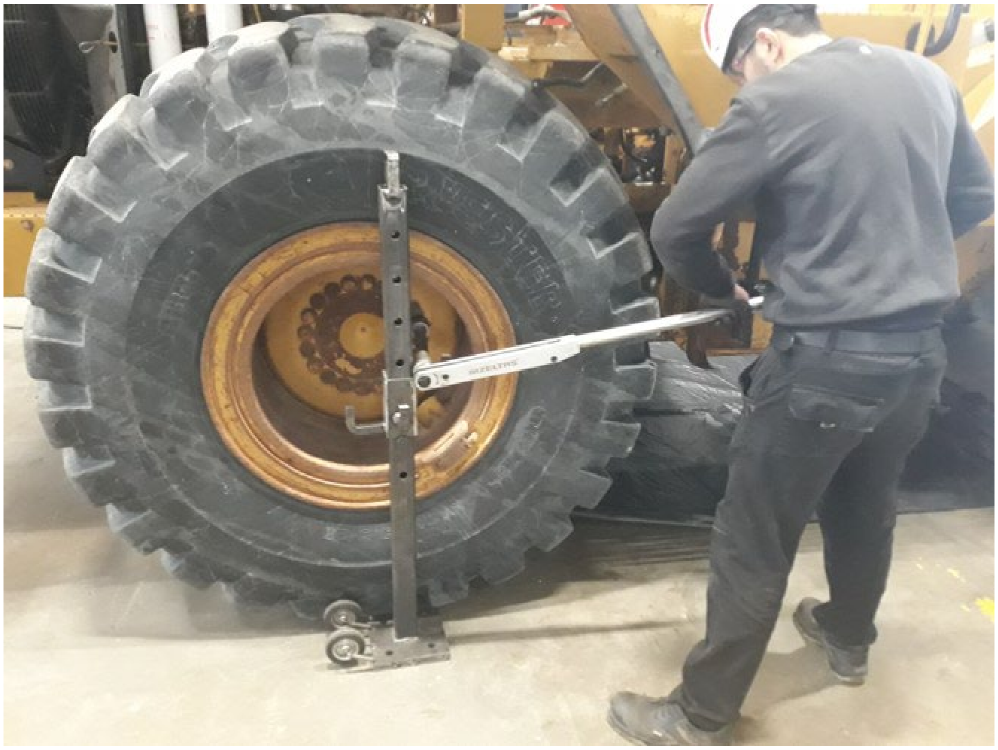

Figure 6: Assembly of the lug nuts with the new apparatus

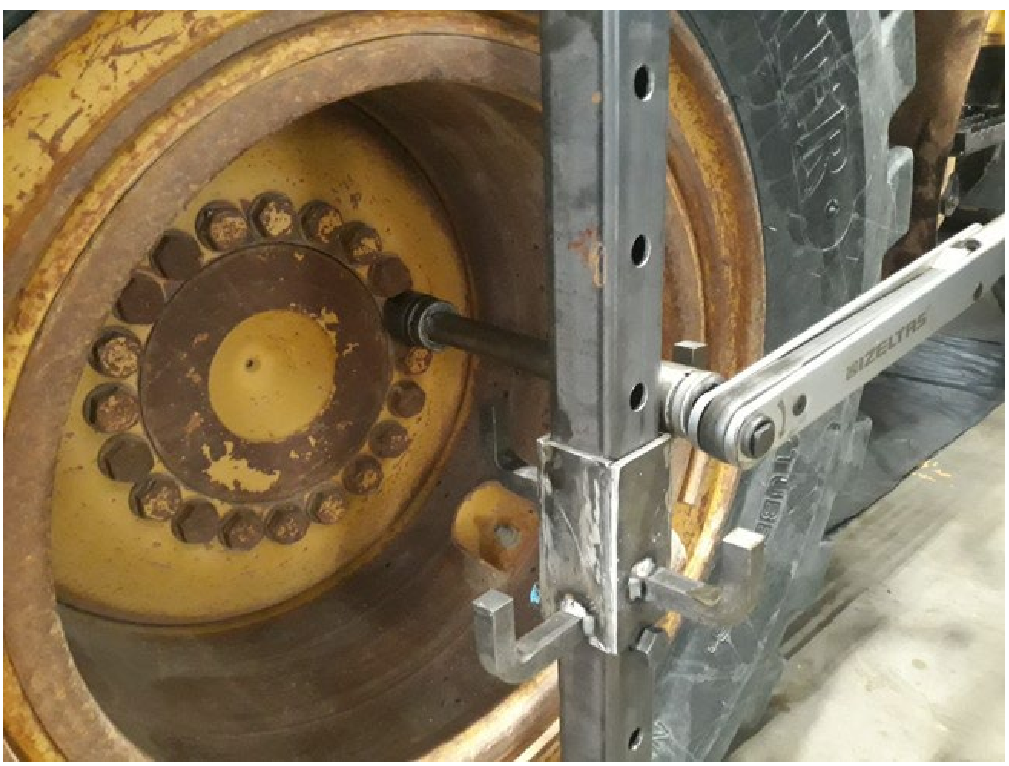

Figure 7: Various support places located on the apparatus

Figure 8: Tools are used safely, parallel to the ground

4.0 Benefits

- Diverse range of use, because it can be adjusted according to the machine size and the lug nut height;
- Convenient to use in the field and easy to transport;
- Provides support for the long intermediate arm and socket wrench when removing or tightening lug nuts;
- Allows one person to easily perform the process of removing and tightening the lug nuts;
- Prevents injury or damage by eliminating the risk of the socket wrench or long intermediate arm being dislodged during disassembly and assembly.

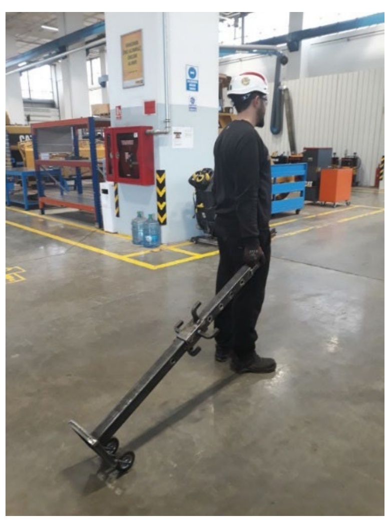

Figure 9: The apparatus can be easily transported

5.0 Resources Required

Item	Cost
Material	$35
Labor	3 hours

6.0 Supporting Attachments / References

- Drawing (attached)

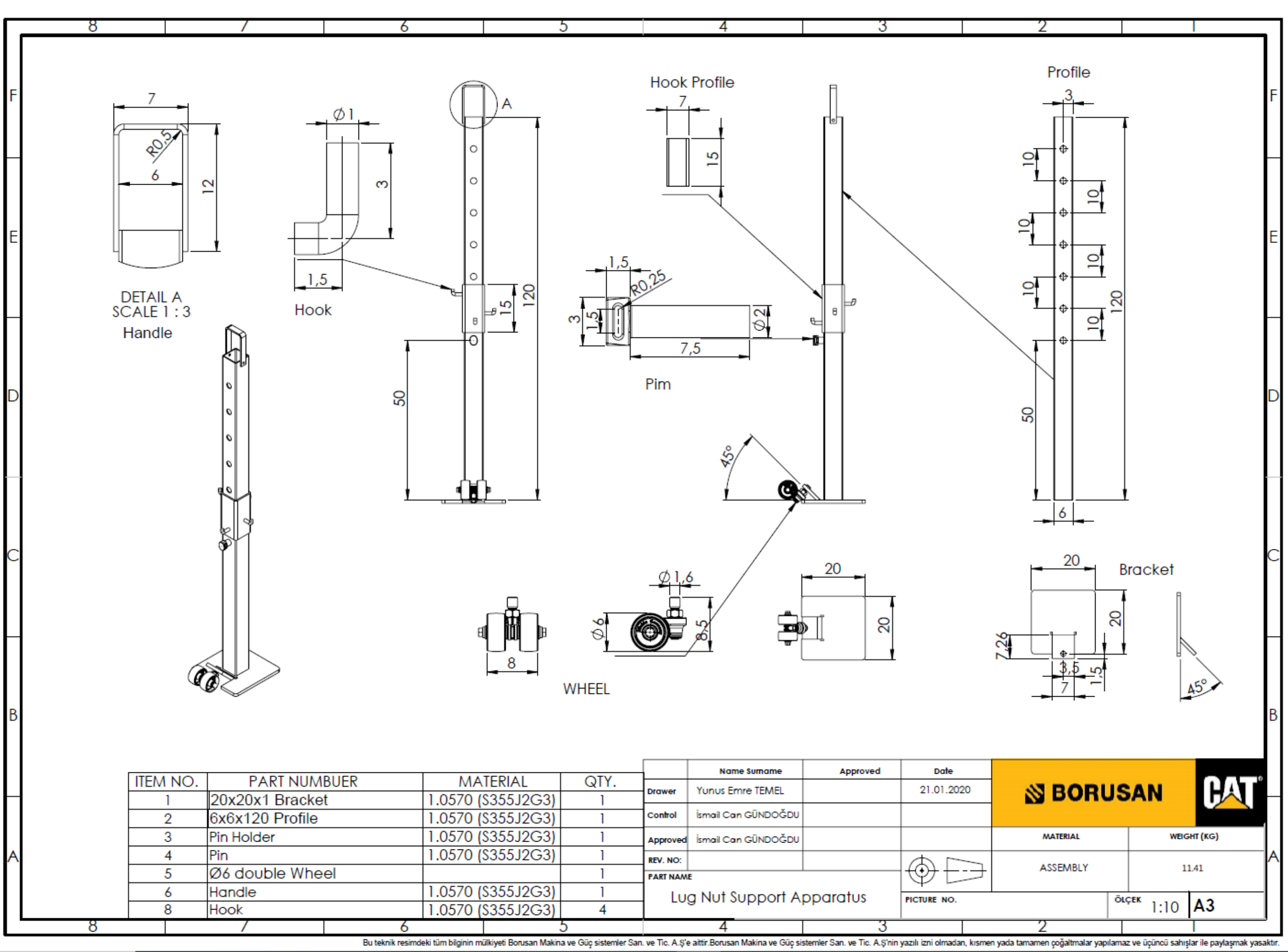

Figure 10: Engineering print of the apparatus. Dimensions in cm

7.0 Related Best Practices

N/A

8.0 Acknowledgements

This Best Practice was written by:

Mustafa Tuzcu
CRC Engineer Intern
Borusan Makina ve Güç Sistemleri

This Best Practice designed by:

Ferdi Yiğit
Quality and CI Specialist
Borusan Makina ve Güç Sistemleri

Project and Best Practice supported by:

Barış Kaan Bilir
Ankara CRC Team Leader
Borusan Makina ve Güç Sistemleri

Volkan Yayman
CRC Manager
Borusan Makina ve Güç Sistemleri